Milad Azami

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Materials Science & Engineering professional with extensive experience in material characterization and hands-on experience in a high volume manufacturing environment. Skilled in lean six sigma, failure root cause analysis, process development, optimization, and documentation, statistical data analysis, and cross-functional collaboration.

Technical Skills

- Project Management: Team Collaboration, Vendor Coordination, Documentation and SOPs.
- Process Development&Optimization: Quality Control, Risk Management, Real-time Data Monitoring.
- Tool Ownership: Tool Qualification, Error Recovery, Failure Root Cause Analysis.
- Materials Testing and Characterization: XRD, SEM, TEM, EDS, FIB, TGA, EDS, FTIR, XPS, SIMS.
- Li ion Battery Electrode Processing and Battery Cell Assembly : LTO, NCM, LFP, LCO, Electrolytes, Slurry Formulation, Coating, Calendaring, and Drying, Cell Manufacturing.
- Battery Cell Diagnostic/Testing: CV, EIS, Cycle Life, Rate Capability Testing.
- Battery Modeling: COMSOL MULTIPHYSICS, PYTHON.
- Statistical Data Analysis: JMP, SQL.

Experience

Process Engineer / Intel, OR (Mar 2022 – Sep 2025)

- Proficient in failure root cause analysis, tool recovery, and disposition.
- Developed and implemented standard operating procedures for tool and process recovery.
- Authored white papers documenting process improvements, pilot results, and defect mitigation strategies.
- Managed RFCs and BKMs for tool error recovery (e.g., robot timeouts, bath fails), improving system uptime.
- Eliminated recurring Cu/Ni delamination defects by DOE to pinpoint the failure root cause to stray current and optimized soaking time, resulting in complete elimination of crack and increasing of production yield.
- Collaborated with vendors and FSEs to ensure on-spec performance.
- Collaborated on development of new criteria to catch real time tool marginalities before tool failures.
- Developed and implemented data-driven monitoring systems that became standard practice for tool health tracking.
- Collaborated with NPI team for development of plating recipes.

Battery Scientist / Moses Lake Industries, OR (Dec 2020 – Mar 2022)

- Conducted binder screening and process optimization to improve electrode quality and manufacturability.
- Designed and executed test protocols for electrochemical evaluation and quality control.
- Operated and maintained advanced electrode fabrication and testing setups (coin cell, Swagelok).
- Established diagnostic metrics and metrology standards for electrode performance analysis.
- Collaborated with internal and external teams to streamline manufacturing workflows and material qualification.

Postdoctoral Fellow / **Clemson University, Clemson, SC** (May 2020 – Dec 2020)

- Investigated failure modes, mechanism and analysis (FMMEA) of NMC811 half-cell and full-cell systems through various analytical methods including pOCV, electrochemical cycling, EIS, and IC-DV techniques.
- Analyzed degradation impact of electrolyte additives.

Research Assistant, PhD / **Clemson University, Clemson, SC** (Jan 2015 – Apr 2020)

- Investigated and illustrated the effect of MoO3 dopant level on transport properties of Lithium Titanate (LTO) and proposed novel compensation mechanism using XPS and electronic/ionic conductivity measurements.
- Illustrated the effect of MoO3 dopant level and processing parameters on structure and microstructure of LTO using XRD, EDX, TGA, RAMAN, TGA, SEM, TEM, and BET.
- Conducted battery manufacturing, formulation and design like Coin and Swagelok cells.
- Conducted benchmark studies of materials, additives, and solvents, and provided assessment of their potential through analytical and electrochemical testing.
- Evaluated battery cells through various characterization and data analyses techniques such as cyclic voltammetry (CV), impedance spectroscopy (EIS), Cycle life, Rate capability, etc.
- Developed an elegant, novel and practical processing technique by combining tape casting with freeze casting to design an electrode microstructure with ordered macro porosity channels to facilitate electrolyte transport in ultra-thick electrodes for high power/energy density Li ion batteries.
- Designed experiments to obtain electrochemical parameters of including electrode and separator tortuosity, OCV and Li ion diffusivity as a function of SOC, and electrode reaction rate.
- Managed different project and lead different groups, leading to publishing of 4 reviewed journals

Education

Ph.D. in Materials Science and Engineering / **Clemson University, Clemson, SC** (Jan 2015 – May 2020)

Dissertation: Engineered Porous Electrode for High Performance Li-ion Batteries. Among the top 10% of PhD students, Graduated with GPA = 4.0. Bishop Fellowship recipient (2019-2020). Postdoctoral Fellowship recipient.

MSC. in Materials Science and Engineering / **Sharif University of Technology, IR** (Sep 2010 – Jan 2013)

Dissertation: Thesis: Synthesis of CNT/CNF Nanofibers Through Electrospinning

Publications (Selected)

- Azami, M., et al. “Effect of isotropic and anisotropic porous microstructure on electrochemical performance...” Journal of Power Sources, 2020.
- Azami, M., et al. “Novel insights into $Li_4Ti_5O_{12}$ transport properties and air stability.” Energy Storage Materials, 2020.
- Maiti, S.C., Azami, M., et al. “Dopants on Ion Transport: Mo^{6+}-doped Lithium Titanites.” Ceramic International, 2018.